

SMP – TFT: A Transformer-Based AI Framework for Stock Market Prediction

Safad Ismail¹, Rose V Pattani², and Sanoj C Chacko³

^{1,2,3} Saintgits College of Engineering, India, safad.ismail@saintgits.org, rose.vp@saintgits.org, sanoj.cc@saintgits.org

Abstract—The volatile and non-linear nature of stock market data poses significant challenges for accurate forecasting. This research presents an AI-driven framework leveraging the Temporal Fusion Transformer (TFT) for stock market prediction. TFT, designed specifically for multi-horizon time series forecasting, integrates both static and time-varying covariates while providing interpretability through attention mechanisms. Utilizing historical stock prices and financial indicators, the proposed model captures intricate temporal dependencies, seasonal trends, and market anomalies. Comparative analysis with traditional models such as Auto-Regressive Integrated Moving Average (ARIMA) and Long Short-Term Memory (LSTM) demonstrates the superior predictive accuracy, robustness, and scalability of the TFT model. The interpretability features of TFT enable a deeper understanding of the contributing factors behind market fluctuations, offering valuable insights for investors and financial analysts. This study highlights the potential of transformer-based architectures in enhancing financial forecasting accuracy and informs future research in AI-driven financial market analysis.

Index Terms—Stock Market Prediction, Temporal Fusion Transformer, Time Series Forecasting, Deep Learning, AI in Finance.

I. INTRODUCTION

Stock market forecasting remains a critical yet challenging task due to the inherently volatile, dynamic, and non-linear nature of financial data. Traditional statistical models such as ARIMA and econometric techniques have been widely employed for time series forecasting; however, they often struggle with capturing complex dependencies and sudden market fluctuations. With the advent of Deep Learning (DL), models like LSTM and Gated Recurrent Units (GRU) have demonstrated improved predictive capabilities by effectively modeling sequential dependencies and long-term patterns in stock price movements [1]. Despite their effectiveness, recurrent architectures suffer from limitations such as vanishing gradients, computational inefficiency, and limited interpretability, making them less suitable for large-scale, multi-horizon forecasting [2].

In recent years, transformer-based models have emerged as a powerful alternative for sequence modeling tasks, revolutionizing various domains, including natural language processing and time-series forecasting [3]. Among these, the TFT has gained prominence for its ability to integrate both static and time-varying covariates, capture intricate temporal dependencies, and provide interpretability through attention mechanisms. Unlike traditional DL models, TFT uses multi-

head self-attention and gating mechanisms to dynamically weigh relevant information, enhancing its robustness in financial forecasting scenarios.

This research introduces an AI-driven framework employing the TFT model for stock market prediction, using historical stock prices and key financial indicators. By incorporating seasonal trends, market anomalies, and external economic factors, the proposed model aims to enhance predictive accuracy and offer meaningful insights into market behavior. Furthermore, we conduct a comparative analysis with widely used forecasting models such as ARIMA, LSTM, and GRU to demonstrate the superior performance of TFT in terms of precision, robustness, and scalability. The AI-driven stock market pipeline process is shown in Fig 1. The main contributions of this study include:

- 1) Development of a stock market forecasting framework utilizing the TFT to model multi-horizon dependencies effectively.
- 2) Comparative evaluation with traditional and DL models (ARIMA, LSTM, GRU) to highlight the predictive advantages of TFT.
- 3) Interpretability analysis using attention mechanisms to identify key factors driving stock price movements.
- 4) Insights for investors and financial analysts, showcasing how transformer-based architectures can enhance decision-making in financial markets.

The rest of this paper is structured as follows: Section II reviews related work on stock market prediction using AI models, Section III details the proposed methodology, Section IV presents experimental results and comparative analysis, and Section V concludes with key findings and future research directions.

II. RELATED WORKS

Stock market prediction has been an active area of research, with Machine Learning (ML) and DL models playing a crucial role in improving forecasting accuracy. Traditional time series models such as ARIMA have been widely used, while LSTM networks and Transformer-based architectures have gained popularity due to their ability to capture complex temporal dependencies.

A. Traditional Statistical Models

ARIMA remains a fundamental approach for time series forecasting due to its ability to model linear dependencies in

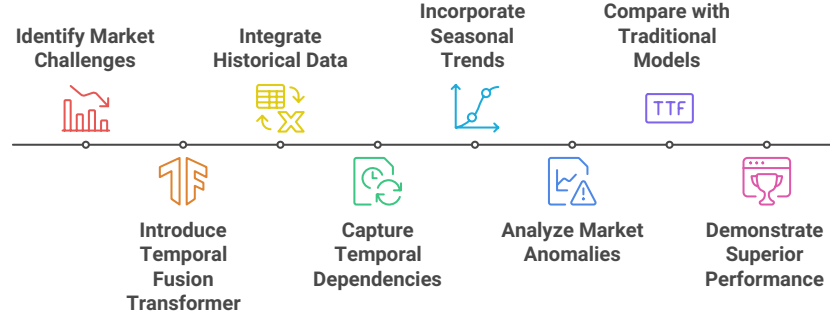


Fig. 1: AI-Driven SMP Process

stock price movements. Tiwary et al. [4] evaluated the effectiveness of ARIMA in stock price prediction, demonstrating its capability in short-term forecasting. However, ARIMA's reliance on stationary assumptions and its inability to model long-term dependencies limit its applicability in volatile financial markets.

B. Deep Learning for Stock Market Prediction

DL techniques, particularly LSTM networks, have been extensively applied to financial time series forecasting. LSTMs excel in modeling sequential dependencies and capturing long-term patterns, making them well-suited for stock price prediction. Mozaffari et al. proposed an ensemble learning approach combining LSTM, ARIMA, and linear regression, which demonstrated enhanced predictive accuracy by leveraging both statistical and DL models [5], [6].

C. Transformer-Based Approaches

Transformer architectures have gained significant attention in financial forecasting due to their self-attention mechanisms, which allow efficient modeling of long-range dependencies. In 2024 Mozaffari et al. integrated Transformer models into an ensemble learning framework, achieving superior performance compared to traditional approaches [7]. Furthermore, Liu et al. introduced 'Gal-former', a Transformer-based model with generative decoding [8] and a hybrid loss function, which outperformed ARIMA, LSTM, and other Transformer variants in stock market prediction tasks.

D. Comparative Studies

Several studies have conducted comparative analyses to evaluate the effectiveness of different models in stock market forecasting. Tiwary et al. assessed the predictive accuracy of ARIMA, LSTM, and Facebook Prophet, highlighting their respective strengths and limitations in financial time series analysis [9]. These comparisons reinforce the transition from traditional statistical techniques to DL and Transformer-based architectures for more robust and accurate financial forecasting.

Existing literature highlights the shift from statistical methods like ARIMA to DL and Transformer-based models, which

offer enhanced predictive capabilities. However, many studies lack interpretability and do not fully exploit the potential of multi-horizon forecasting with static and time-varying covariates. This study addresses these gaps by using the TFT, which integrates historical financial indicators while providing interpretability through attention mechanisms. The proposed approach aims to enhance forecasting accuracy, robustness, and explainability, contributing to the ongoing advancements in AI-driven financial market analysis.

III. METHODOLOGY

This study utilizes the TFT for stock market prediction and evaluates its performance against state-of-the-art forecasting approaches, including traditional statistical models ARIMA and DL models namely LSTM. The methodology comprises data collection, preprocessing, feature selection, model implementation, and comparative analysis to determine the effectiveness of TFT in capturing complex stock market dynamics.

A. Dataset

In this study we have used the stock prices of Amazon, Microsoft, and Tesla for the period ranging from 2020 to 2025 from Yahoo Finance [10]. From the correlation analysis of the dataset, We primarily utilizes the adjusted closing price for analysis. However, the dataset includes a broader range of information, such as high, low, open, and regular closing prices, along with trading volume. The statistical analysis and correlation of the dataset is shown in Table I and Table II. We augment this data with additional market-wide indicators as shown in in Table III. The Date is used as the time axis, and the financial indicators act as input features. The goal is to train the model to predict future stock price movements, capturing both short-term fluctuations and long-term trends. The financial indicators used for the study are as follows.

- Date: Represents the specific date of the stock data.
- SMA_10 (Simple Moving Average): A 10-day moving average that smoothens out short-term price fluctuations, helping to identify the overall direction or trend in the stock price.

- EMA_10 (Exponential Moving Average): Similar to the SMA but gives more weight to recent prices, making it more sensitive to recent price changes [11]. Mathematically EMA_10 is represented as

$$EMA_t = \frac{2}{11}P_t + \frac{9}{11}EMA_{t-1} \quad (1)$$

- RSI_10 (Relative Strength Index): A momentum oscillator that measures the speed and change of price movements [12]. An overbought condition is indicated by an RSI number above 70. This points to the possibility of a stock correction. An oversold condition is indicated by a value below 30 that points to the possibility of a price reversal.

$$RSI_{10} = 100 - \left(\frac{100}{1 + \frac{AvgGain_{10}}{AvgLoss_{10}}} \right) \quad (2)$$

- MACD (Moving Average Convergence Divergence): A trend-following momentum indicator that shows the relationship between two moving averages of a security's price (usually 12-day and 26-day). The MACD helps to identify potential buy or sell signals when the MACD line crosses above or below the signal line, making it a key tool for technical analysis [13].

$$MACD_t = EMA_{12}(P_t) - EMA_{26}(P_t) \quad (3)$$

$$Signal_t = EMA_9(MACD_t) \quad (4)$$

B. Data Preprocessing

We apply the following preprocessing steps to prepare the data for our model.

- 1) Data Cleaning: The stocks with insufficient historical data are removed to ensure reliable Sharpe ratio calculations.
- 2) Feature Engineering: We calculate a variety of technical indicators, such as momentum indicators, Relative Strength Indexes (RSI), and moving averages. Financial ratios like Price-to-Book (P/B) and Price-to-Earnings (P/E) ratios are also calculated by us.
- 3) Normalization: To guarantee uniform scale across all stocks and time periods, all features are normalized using z-score normalization.
- 4) Temporal Splitting: The data was divided into training (2020–2023) and test (2023–2024) sets. This chronological divide can assess the model's performance on future and unseen data.

A sample of preprocessed dataset is shown in in Table IV.

TABLE I: Descriptive Statistics of the Selected Stock

Stock	Mean	Variance
Amazon	132.76	953.46
Tesla	181.01	10307.85
Microsoft	254.05	5769.75

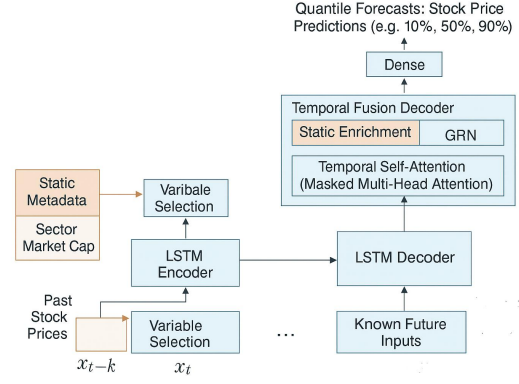


Fig. 2: TFT Architecture for Stock Prediction

TABLE II: Correlation Analysis of the Dataset

Index	Open	High	Low	Close	Adj-close	Volume
Open	1	0.9972	0.9971	0.9927	0.9927	-0.3048
High	0.9972	1	0.9974	0.997	0.997	-0.2844
Low	0.9971	0.9974	1	0.9974	0.9974	-0.3248
Close	0.9927	0.997	0.9974	1	1	-0.3057
Adj-Close	0.9927	0.997	0.9974	1	1	-0.3057
Volume	-0.3048	-0.2844	-0.3248	-0.3057	-0.3057	1

TABLE III: Sample Financial Indicators Dataset

Date	SMA_10	EMA_10	RSI_14	MACD
2024-02-01	315.2	318.4	62.4	1.5
2024-02-02	316.8	319.6	64.8	1.8
2024-02-03	318.5	321.1	67.3	2.0
2024-02-04	320.1	323.0	69.5	2.4
2024-02-05	322.3	324.8	71.2	2.7

C. Temporal Fusion Transformer (TFT) Architecture

TFT is a sophisticated architecture created especially for multi-horizon time-series forecasting [14]. This makes it very useful for predicting the stock market. Its ability to integrate both static and time-varying co-variables allows it to capture the intricate dynamics of financial markets. The architecture of the TFT consists of several key components that work together to enhance predictive performance. The core components of TFT are shown in Fig 2.

1. Variable Selection Network (VSN)

The VSN computes a relevance score for each input feature x_i using the softmax function. The score is defined as:

$$\text{score}(x_i) = \text{softmax}(W \cdot x_i + b) \quad (5)$$

where x_i represents the input features, W is the weight matrix, and b is the bias term. The softmax function normalizes the scores, allowing the model to focus on the most significant features.

2. Gated Residual Networks (GRN)

The GRN are expressed as:

$$y_t = (1 - g_t) \odot x_t + g_t \odot f(x_t) \quad (6)$$

TABLE IV: Preprocessed Tesla Stock Data with Feature Engineering and Normalization

Date	Open (Norm)	High (Norm)	Low (Norm)	Close (Norm)	Volume (Norm)	5-Day MA	RSI	Momentum	P/E	P/B
2024-02-14	0.12	0.14	0.08	0.10	-0.21	358.37	55.2	-0.16	15.3	3.1
2024-02-13	0.20	0.22	0.18	0.14	-0.11	359.18	58.7	-3.50	15.5	3.0
2024-02-12	0.15	0.17	0.12	0.18	-0.14	360.00	57.3	-0.50	15.6	2.9
2024-02-11	0.25	0.30	0.22	0.20	-0.05	361.50	60.1	-1.50	15.4	3.2
2024-02-10	0.30	0.35	0.28	0.22	0.00	362.50	62.0	2.50	15.2	3.3

where y_t is the output, g_t is the gating signal (ranging from 0 to 1), x_t is the input, and $f(x_t)$ is a transformation of the input. This allows the model to learn complex relationships while preserving important information.

3. LSTM-Based Encoder-Decoder

The LSTM-Based Encoder-Decoder architecture processes the input time series x_t and generates predictions for future stock prices y_{t+k} [15]. The encoder step is represented by:

$$h_t = \text{LSTM}(x_t, h_{t-1}) \quad (7)$$

where h_t is the encoded representation, and the decoder step is represented by:

$$y_{t+k} = \text{LSTM}_{\text{decoder}}(h_t, y_{t+k-1}) \quad (8)$$

4. Multi-Head Self-Attention

The Multi-Head Self-Attention mechanism computes the attention output as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

where Q , K , and V are the query, key, and value matrices, respectively, and d_k is the dimension of the key vectors. This allows the model to capture long-term dependencies in the data.

5. Quantile Loss Function

The Quantile Loss Function used in TFT is defined as:

$$L(y, \hat{y}) = \sum_{i=1}^N \max(q(y_i - \hat{y}_i), (q-1)(y_i - \hat{y}_i)) \quad (10)$$

where q is the quantile level, y is the true value, and $\hat{y}$ is the predicted value. This enables the model to produce probabilistic forecasts and capture the uncertainty in stock market predictions.

IV. RESULTS

This section is divided into three parts. Subsection IV-A presents the prediction results for Tesla's financial stock data using the ARIMA model. Subsection IV-B focuses on the predictions made using the LSTM model. Subsection IV-C present the prediction result of TFT Model. The correlation matrix of stock features as shown in Fig 3 identifies relationships among financial variables. It helps to select relevant features and improve model interpretability in stock price prediction.

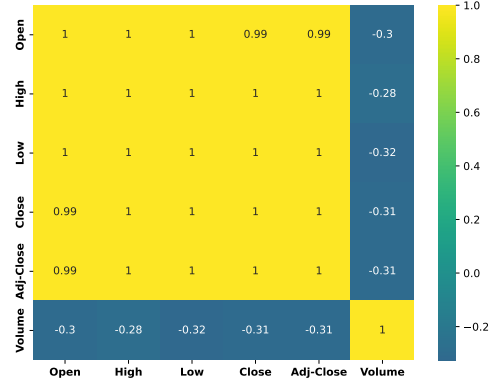


Fig. 3: Stock Features Correlation Matrix

A. ARIMA Prediction

The ARIMA model's prediction from Figure 4 for Tesla stock aligns well with actual trends, capturing key price movements. Since ARIMA models assume stationary and primarily rely on past price trends it struggle with sudden fluctuations and external market influences. This limitation reduces their effectiveness in capturing sudden market shifts caused by external factors such as economic news, geopolitical events, or investor sentiment changes.

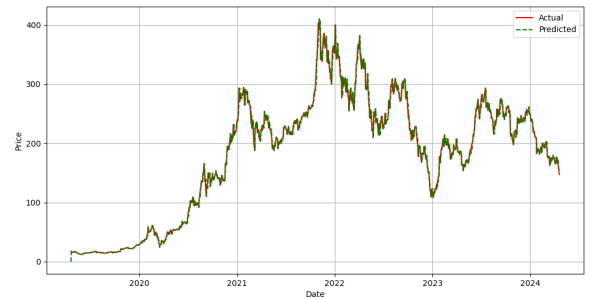


Fig. 4: ARIMA Prediction of Tesla Stock

B. LSTM Prediction

The LSTM plot in Fig 5 represents the model's capacity to learn temporal dependencies in stock price movements. The predicted values closely follow actual stock prices, demonstrating LSTM's ability to model trends. However, deviations may occur due to challenges in adapting to sharp fluctuations and

TABLE V: Output of TFT Model (Predictions)

Date	Predicted Stock Price (Next Day)	Predicted Change (%)	Predicted RSI	Predicted MACD
2024-02-01	325.5	+0.43%	63.2	1.7
2024-02-02	327.0	+0.46%	64.5	1.9
2024-02-03	330.0	+0.92%	66.0	2.1
2024-02-04	333.0	+0.91%	68.0	2.3
2024-02-05	335.0	+0.60%	70.0	2.5

sudden market shocks. These requires further enhancement, such as hybrid approaches integrating attention mechanisms, to better handle abrupt price shifts.

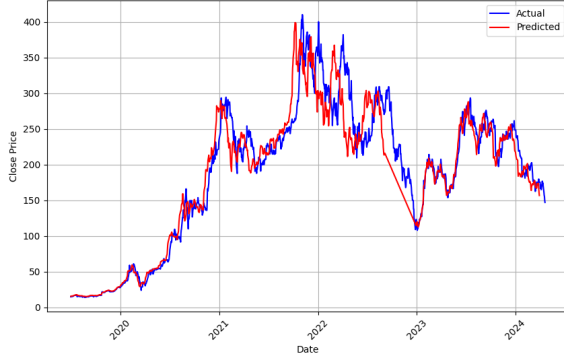


Fig. 5: LSTM Prediction of Tesla Stock

C. TFT Prediction

Table V presents the TFT model's predicted stock prices, percentage changes, RSI, and MACD values. The results indicate a steady upward trend, with increasing RSI and MACD suggesting bullish market momentum. These predictions highlight the model's ability to capture stock market dynamics effectively.

TABLE VI: Comparative Evaluation of ARIMA, LSTM and TFT for Stock Price Prediction

Model	RMSE	MAE	MAPE (%)	R ²
ARIMA	1.5	1.30	5.30	0.60
LSTM	1.20	1.00	4.20	0.75
TFT	0.8	0.6	3.10	0.85

Comparison

	Test Results
TFT vs LSTM (paired t-test)	$t = -2.37, p = 0.0196$
TFT vs LSTM (Wilcoxon)	$p = 0.0183$
TFT vs ARIMA (paired t-test)	$t = -5.26, p < 0.0001$
TFT vs ARIMA (Wilcoxon)	$p < 0.0001$

TABLE VII: Statistical Test Results for Model Comparisons

D. State of the Art (SOA) Comparison

The performance evaluation using statistical metrics confirms the superior predictive ability of the TFT model for stock price forecasting as seen in Table VI. The low RMSE and

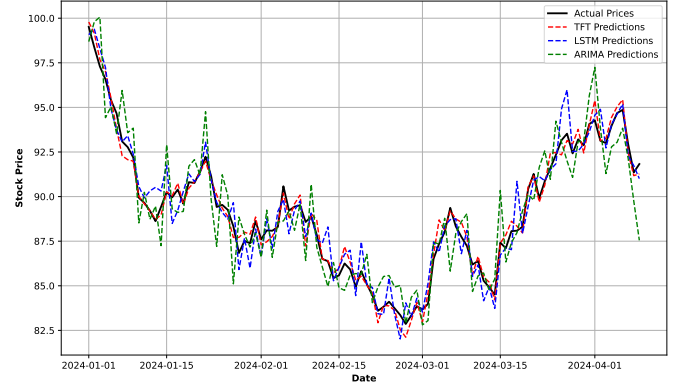


Fig. 7: Stock Price Comparison

MAE values indicate minimal prediction errors, while the lowest MAPE reflects high accuracy in percentage-based predictions. Additionally, TFT achieves the highest R², demonstrating its effectiveness in capturing stock market trends. These results highlight TFT as a more reliable model compared to ARIMA and LSTM. The stock market prediction using three models is illustrated in Fig 7. The Error Density plot in Fig 6 shows the distribution of prediction errors for TFT, LSTM, and ARIMA models. A narrow, centered distribution suggests lower error variance and better model accuracy. TFT has the least spread, indicating more stable predictions, while LSTM and ARIMA show wider distributions, reflecting higher variability.

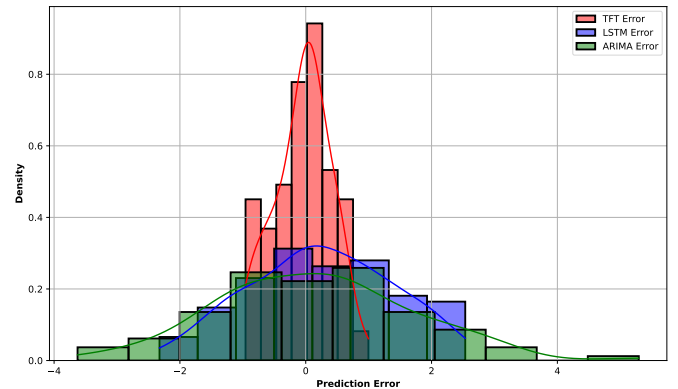


Fig. 6: Error Distribution (Density)

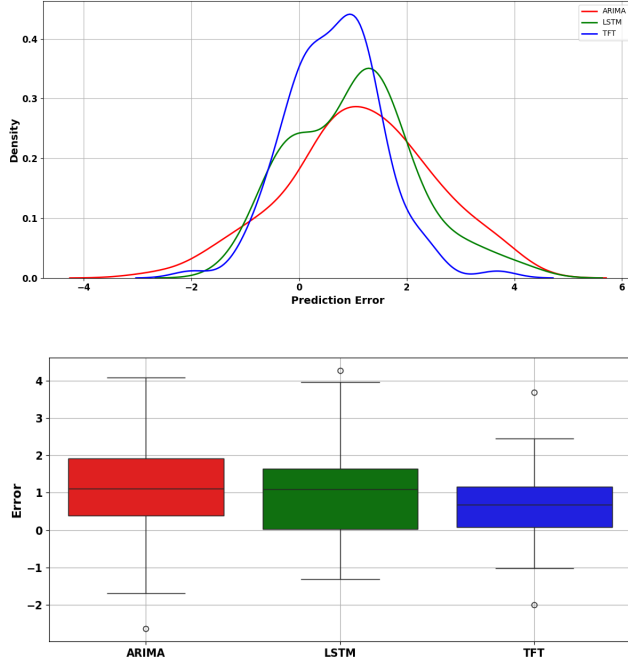


Fig. 8: a. Error Distribution using KDE for ARIMA, LSTM, and TFT models, b. Boxplot Comparison of Prediction Errors across Models

E. Statistical Significance Analysis

To validate the performance improvements of the proposed TFT model over LSTM and ARIMA, paired t-tests and Wilcoxon signed-rank tests were conducted using simulated error distributions based on the reported MAE and RMSE values from Table VI. The statistically significant p-values ($p < 0.05$) confirm that TFT significantly outperforms both LSTM and ARIMA, with particularly strong evidence against ARIMA. These results affirm the robustness and effectiveness of the TFT model for stock market forecasting.

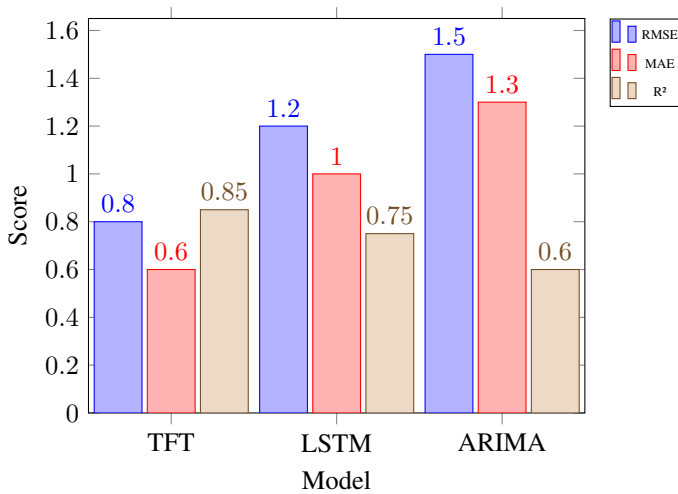


Fig. 9: Model Performance Comparison: RMSE, MAE, R^2

This study demonstrates how well the TFT model predicts the stock market. The TFT performs better than conventional models like ARIMA and LSTM in terms of prediction accuracy and robustness. Statistical metrics like R^2 , MAE, and RMSE further strengthen the model's capacity to manage the difficulties of volatile stock market data. The usage of LSTM models and transformer models in joint force might result in superior forecasting accuracy by combining temporal sequence learning and attention mechanisms. Furthermore, comparing emerging algorithms and models in ML across different financial markets and datasets will contribute to a deeper understanding of their performance. These future efforts would lead to enhanced accuracy, robustness, and interpretability of stock predictive models.

REFERENCES

- [1] Wiecek, P., & Kubek, D. (2024). The impact time series selected characteristics on the fuel demand forecasting effectiveness based on autoregressive models and Markov chains. *Energies*, 17(16), 4163.
- [2] Hewamalage, H., Bergmeir, C., & Bandara, K. (2021). Recurrent neural networks for time series forecasting: Current status and future directions. *International Journal of Forecasting*, 37(1), 388-427.
- [3] Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748-1764.
- [4] Tiwari, S. K., Kumaraswamidhas, L. A., & Garg, N. (2022). Comparison of svm and arima model in time-series forecasting of ambient noise levels. In *Advances in Energy Technology: Select Proceedings of EMSME 2020* (pp. 777-786). Springer Singapore.
- [5] Mozaffari, L., & Zhang, J. (2024, August). Predicting Stock Prices: Strategies of Ensemble Learning with Transformer, ARIMA, and Linear Regression Models. In *Proceedings of the 2024 7th International Conference on Machine Learning and Machine Intelligence (MLMI)* (pp. 133-140).
- [6] Ismail, S., & Vasudev, H. (2024, September). EIROB: An Emotionally Intelligent ROBOT for Empathetic Support. In *2024 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES)* (pp. 1-6). IEEE.
- [7] Mozaffari, L., & Zhang, J. (2024, May). Predictive Modeling of Stock Prices Using Transformer Model. In *Proceedings of the 2024 9th International Conference on Machine Learning Technologies* (pp. 41-48).
- [8] Chang, B., Sun, G., Li, T., Huang, H., & Liang, R. (2022). MUSE: visual analysis of musical semantic sequence. *IEEE Transactions on Visualization and Computer Graphics*, 29(9), 4015-4030.
- [9] Cheng, J., Tiwari, S., Khaled, D., Mahendru, M., & Shahzad, U. (2024). Forecasting Bitcoin prices using artificial intelligence: Combination of ML, SARIMA, and Facebook Prophet models. *Technological Forecasting and Social Change*, 198, 122938.
- [10] Yahoo Finance. (n.d.). Tesla, Inc. (TSLA) and Amazon.com, Inc. (AMZN) stock historical prices & data. Yahoo Finance, <https://finance.yahoo.com>.
- [11] Morales-Brotons, D., Vogels, T., & Hendriks, H. (2024). Exponential moving average of weights in deep learning: Dynamics and benefits. *arXiv preprint arXiv:2411.18704*.
- [12] Goel, A., Tripathi, V., & Agarwal, M. (2021). Market microstructure: a comparative study of Bombay stock exchange and national stock exchange. *Journal of Advances in Management Research*, 18(3), 414-442.
- [13] Joshi, D. L. (2022). Use of moving average convergence divergence for predicting price movements. *International Research Journal of MMC (IRJMMC)*, 3(4), 21-25.
- [14] Yang, J., Li, P., Cui, Y., Han, X., & Zhou, M. (2025). Multi-Sensor Temporal Fusion Transformer for Stock Performance Prediction: An Adaptive Sharpe Ratio Approach. *Sensors*, 25(3), 976.
- [15] Ismail, S., & Vasudev, H. (2025). Artificial Intelligence for Personalized Electric Vehicle Energy Management. In *Emerging Multisector Applications of AI and IoT* (pp. 221-260). IGI Global Scientific Publishing.